Digital Buck-Boost (LTC3440): Working Document

Digital Audio Player • Rev 7

Author: Alexander Loker. Design decisions, component selection, analysis and verification.

Drafted by: Claude (Anthropic). Structure and prose from the author's direction and source data.

Contents

1. Summary 1

2. Design Requirements 1

3. Pin Configuration 2

4. External Components 2

5. Equations and Results 2

6. Important Values 3

7. Open Problems 3

8. Revision Log 3

9. Sources 3

1. Summary

Field	Value
Manufacturer / MPN	Analog Devices, LTC3440EMS#TRPBF (#1-050), U205
Package	10-lead MSOP (MS)
Function	Synchronous buck-boost, SYS → +3V3_DIG

2. Design Requirements

Requirement	Value
Input	SYS, 3.0 to 4.2 V
Output	3.3 V
Load	ESP32-S3, flash, microSD, display logic, DAC digital, pull-ups
Output current	300 mA max (≈ 260 mA S4 peak)
Switching frequency	600 kHz (0.48 to 0.72 MHz), separated from the analog LTC3440
Mode	Fixed frequency (MODE/SYNC to GND), always on
Constraints	Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1	RT	R_T to GND
2	MODE/SYNC	GND (fixed frequency)
3, 4	SW1, SW2	L
5	GND	GND
6	VOUT	+3V3_DIG, C_OUT (3 ×), R1, C_FF
7	VIN	SYS, C_IN
8	SHDN/SS	R_SS from SYS, C_SS to GND (always on, soft-started)
9	FB	R1 / R2 midpoint, C_FF
10	VC	R_Z in series with C_P1 to FB

Source: [2]

4. External Components

Ref	Value / Rating	MPN	AL-PN
L (L204)	18 µH ±20 %, I_sat 1.8 / 2.8 / 3.7 A, 62.5 mΩ	Coilcraft XGL5050-183MEC	#1-094
C_IN (C224)	10 µF 10 V X7R 0805	Murata GCM21BR71A106KE22K	#1-081
C_IN HF (C234)	0.1 µF 50 V X7R 0402	Murata GCM155R71H104KE02D	#1-080
C_OUT (C227, C228, C229)	3 × 22 µF 25 V X7S 1210	Murata GCM32EC71E226KE36L	#1-073
R1 (R236)	172 kΩ 0.1 % 0603	YAGEO RT0603BRD07172KL	#1-095
R2 (R240)	100 kΩ 0.1 % 0402	YAGEO RT0402BRD07100KL	#1-131
R_T (R239)	100 kΩ 0.1 % 0402	YAGEO RT0402BRD07100KL	#1-131
R_Z (R238)	49.9 kΩ 0.1 % 0402	YAGEO RT0402BRD0749K9L	#1-138
C_P1 (C226)	1.5 nF 2 % C0G 0402	Murata GRM1555C1H152GA01D	#1-077
C_FF (C225)	100 pF 5 % C0G 0402, across R1	Murata GRM1555C1H101JA01D	#1-096
R_SS (R237)	54.9 kΩ 0.1 % 0402	YAGEO RT0402BRD0754K9L	#1-079
C_SS (C230)	0.1 µF 50 V X7R 0402	Murata GCM155R71H104KE02D	#1-080
Power LED (D209), R253	Green 0603; 1 kΩ 0.1 % 0402	Lite-On LTST-C190GKT; YAGEO RT0402BRD071KL	#1-009, #1-043

Source: [2]

5. Equations and Results

$$R_T = 6 \times 10^{10} / 600\,kHz = 100\,k\Omega; \quad V_{OUT} = 1.22\,(1 + 172/100) = 3.318\,V \tag{1}$$

$$L_{buck} > \frac{3.3 \times 0.9}{0.6 \times 0.3 \times 0.3 \times 4.2} = 13.1\,\mu H \rightarrow 18\,\mu H \tag{2}$$

$$I_{L,\,pk} = \frac{0.3 \times 3.3}{3.0 \times 0.85} + \frac{0.025}{2} = 0.40\,A \tag{3}$$

$$f_{RHPZ} = \frac{V_{IN}^2}{2\pi I_{OUT} L V_{OUT}} = \frac{3.0^2}{2\pi \times 0.3 \times 18\,\mu H \times 3.3} = 80\,kHz \tag{4}$$

$$f_{LC} = \frac{1}{2\pi\sqrt{18\,\mu H \times 66\,\mu F}} = 4.6\,kHz; \quad f_z(R_Z C_{P1}) = 2.1\,kHz \tag{5}$$

$$f_z(R1\,C_{FF}) = 9.3\,kHz; \quad f_p((R1 \| R2)\,C_{FF}) = 25\,kHz \tag{6}$$

$$\tau_{SS} = 54.9\,k\Omega \times 0.1\,\mu F = 5.49\,ms; \quad I_Q = f(0.75V_{IN} + 0.25V_{OUT}) = 2.39\,mA\ (4.2\,V) \tag{7}$$

Source: [1]. f_LC uses nominal C_OUT; DC-bias derating at 3.3 V raises it. Compensation verified by the author.

6. Important Values

Item	Value
Switch current limit	2.7 A typ (2.5 to 2.9 A)
Output	3.318 V
Inductor ripple	21.8 % buck / 8.4 % boost
Gate-charge I_Q	2.39 mA at 4.2 V in (≈ 10 mW); continuous, since the converter is always on
Boost RHP zero	≈ 80 kHz; loop crossover well below
Inductor height	5.10 mm

7. Open Problems

None.

8. Revision Log

Rev	Date	Change
0 to 4	2026-09-23	Initial issue; MODE/SYNC to GND; R1 172 kΩ; inductor.
5	2026-09-25	Reformatted.
6	2026-10-03	Schematic values recorded: SHDN/SS RC soft-start from SYS (R237, C230); compensation R_Z 50 kΩ with C_P1 1.5 nF and 100 pF feed-forward across R1, verified by the author; C_OUT 3 × 22 µF; C_IN 10 µF; R1 #1-095; R_T and R2 0402 #1-131. Designators added.
7	2026-10-16	Datasheet link added to [1]. Reviewed against the schematic: R_Z 49.9 kΩ 0402 (#1-138); C234 0.1 µF input cap and +3V3_DIG power LED (D209, R253) added.

9. Sources

1. LTC3440 Product Page and Datasheet (Analog Devices)

2. DAP schematic sheet DAP-Power-2, 3.3 V buck-boost (author, 2026-10-16)